

SecondSight: Demonstrating Cross-Device Augmented Reality

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ABSTRACT

SecondSight is a framework for rapid prototyping of cross-device interfaces that combine an optical see-through Augmented Reality (AR) head-mounted display (HMD) with a smartphone. SecondSight contains code for simulating AR HMDs with different Fields of View (FOV), providing gesture and head pointing input, and allowing virtual content to be placed in different coordinate frames. Overall, this gives AR researchers flexibility to explore different types of cross-device interfaces and interaction techniques with different types of display, input, and content placement.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Mixed / augmented reality.

KEYWORDS

Augmented Reality, Hybrid Interface, Mobile Phone

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1 INTRODUCTION

SecondSight is a framework for rapid prototyping of cross-device AR interfaces which combine optical see-through AR HMDs and smart phones. Recently companies have begun to develop AR HMDs which can be plugged directly into mobile phones (e.g. Nreal display [Nreal, 2021]). However, there has been little research on how smartphones can be combined with AR HMDs. This is partly because of the complexity of developing cross-device interfaces. SecondSight makes it easy to prototype a variety of AR HMD/handheld display (HHD) cross-device interfaces (see Figure 1). The main novelty is that it enables the exploration of a wider variety of display, interaction and content placement options for cross-devices interfaces than previous research, which are designed for a single HMD (for example the Bishare interface [Zhu and Grossman, 2020]).

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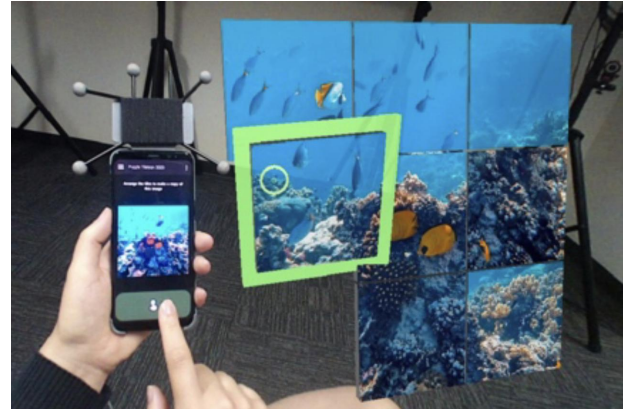
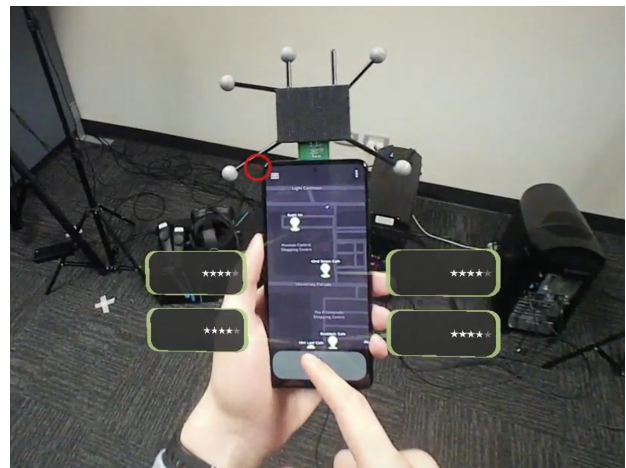


Figure 1: SecondSight applications; (top) Map application, and (bottom) Puzzle game.



The SecondSight platform is a set of Unity packages that can be used to quickly implement different design ideas for using an AR HMD with a smartphone to view and interact with virtual information. The platform has several display components, allowing for virtual information to be shown on the phone screen, the AR HMD in device space view (aligned with the smartphone), and the AR HMD in world space view (aligned with the real world). There are also components for hand tracking and head pointing interaction, and both HMD tracking and phone tracking.

SecondSight can simulate a range of different HMD Fields of View (FOV) to test how AR interface design ideas would be affected by the FOV limitations of current AR headsets. In addition to supporting phone touch-screen input, the system offers two methods to interact

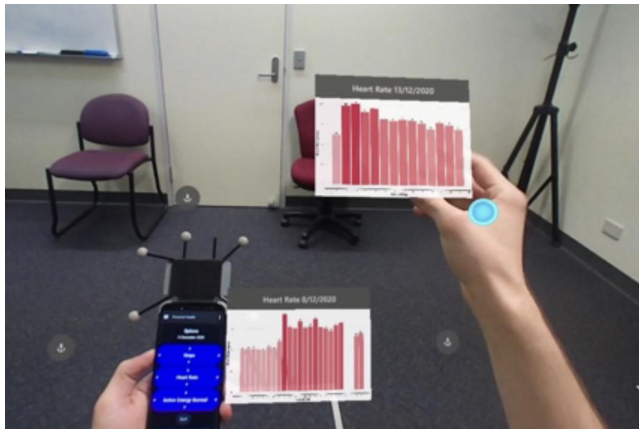


Figure 2: Users can grab a virtual window of a health visualisation application using hand gestures.

with the AR content; free-hand gestures and head-pose. It also allows the placement of virtual AR content in two different ways: 1) World-fixed, where content is placed anywhere in 3D space, and 2) Device-fixed, where the content is aligned on an invisible 2D plane around the phone. Each of these features have a wide range of parameters that can be specified by the interface designer, and so the platform provides a high degree of flexibility in terms of the interface combinations that can be explored.

The system currently supports two modes of interaction with the AR content: Gestures and Head-Pose. These two modes were chosen to enable the assessment of the preferences of a "natural" gesture interaction or using the smartphone for input controls. Currently, the user can select, re-position or delete virtual content around the phone. The action to show virtual content is the same across both interaction modes as it is performed on the phone; the user selects the starting position of the content by swiping the selected content to a desired direction (left, right, up, down).

Using hand gestures, users can grab virtual content to re-position it to a desired location (see Figure 2) or move it to a designated zone below the smartphone, to delete the content. For the option using the Head-Pose, a gaze raycast projects a reticle which hovers over the content. Looking at the content, the user can press a designated area on the bottom of the smartphone. Holding the button on the bottom attaches the virtual object to the user's gaze and allows the user to re-position the object. When the button is released, the content will snap in place. To delete the virtual object, the user looks at an X icon on it and uses their phone to complete the action with a single press.

SecondSight runs on a PC with Windows 10 and the Unity game engine. For the AR prototyping we use the Metavision Meta2 optical see-through AR HMD which has a resolution of 1280×1440 pixels per eye and a wide FoV measuring 82.2°×52°, although other HMDs could also be used. The Meta HMD is connected directly to the PC. We use the Meta SDK for the hand tracking for the gestures. The smartphone used is the Samsung Galaxy S8, although any modern Android phone could be used. The position of the

HMD is tracked in relation to the phone using a ten camera OptiTrack motion capture system [OptiTrack, 2021] with the HMD and smartphone both having OptiTrack tracking balls attached to them. Finally, the communication between devices is enabled through the Mirror Networking API for Unity which enables bi-directional communication between the smart phone and PC/AR HMD.

2 USER EXPERIENCE

The SecondSight framework has been used to create a number of applications, showing different interaction techniques. One of these is a map application where the smartphone is used to view a map view of the user surroundings, while the space around the phone is used to show additional virtual content from the points of interest, viewed on the AR HMD (see Figure 1). When the user taps with their finger on a point of interest a virtual window appears showing more information. The user can either use their natural hands, or head pointing to select and position this window in space, either screen aligned with the phone, or fixed in the real world.

A second application is a simple puzzle (see Figure 1) which allows people to use gestures to arrange virtual tiles around the real phone to complete a picture. Pictures on the real phone screen can be pulled off as virtual images and aligned with the phone. As the user moves the phone around the virtual images move with it. Head pointing or free-hand gestures can be used to manipulate the images, which snap into place when placed next to each other.

A third application is a health visualisation application (see Figure 2), enabling the user to view and compare their personal health data over a period of time. A simple interface is shown on the phone that allows the user to select the day of interest and then select the desired data set from that day that they would like to see. The user can use swipe motions in different directions on the phone screen to create a virtual screen of data aligned with the real phone in that direction. Once the AR graphs are visible they can be arranged in space around the phone, or placed in the world space using hand gestures or head pointing and phone screen input.

This shows how SecondSight makes it easy to develop a range of different intuitive hybrid interfaces that naturally combine phone input and screen viewing with AR content seen on a see-through AR HMD. In the future we will add additional input modalities, such as speech, and conduct user studies to explore which hybrid interaction techniques are best, and the impact of different FOV on performance. Design guidelines from these studies could provide important insights into future AR HMD smartphone experiences.

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